KATHIR V

Bioinformatics Analyst | Protein Sequence Analysis, NGS & Machine Learning

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PROFESSIONAL SUMMARY

Fresher bioinformatics graduate (B.Tech, 2026) with experience in protein sequence analysis, NGS quality control, alignment, variant calling and machine learning. Contributed to a physicochemical k-mer approach that lifted protein family classification from 16.95% to 64.63% accuracy with 74% fewer features. Skilled in Python, Biopython, BLAST and scikit-learn.

TECHNICAL SKILLS

Bioinformatics: Protein Sequence Analysis, k-mer Representation, BLAST, Sequence Alignment, Variant Calling, FastQC, NGS Data Analysis, Biopython

Computational Biology: GROMACS, PyMOL, AutoDock, Physicochemical Feature Engineering

Programming: Python, R, SQL (basic), Linux

Machine Learning: scikit-learn, Random Forest, SVM, KNN, Extra Trees, XGBoost, TensorFlow, PyTorch, NumPy, Pandas

PROJECTS

Physicochemical-Enhanced Embeddings for Protein Family Classification (Team Project) | Python, Biopython, scikit-learn

- Raised accuracy from 16.95% (standard 3-mer baseline) to 64.63%, while cutting features from 1,000 to 261 (74% fewer), by encoding sequences as 3-mers of hydropathy-grouped amino acids on 11,620 UniProt/Swiss-Prot sequences across 10 families.
- Implemented and compared Random Forest, SVM, KNN and Extra Trees, evaluated on an 80:20 stratified split using accuracy, F1-score and confusion matrices.
- Created performance visualizations and optimized the modelling workflow.

NGS Data Analysis (Academic Projects) | FastQC, Sequence Alignment, Variant Calling

- Assessed sequencing read quality using FastQC, then performed sequence alignment and variant calling in academic NGS workflows.

Safe Breath: Lung Sound Classification | Python, Librosa, scikit-learn | Google Play: play.google.com/store/apps/details?id=com.gs.safebreath

- Classified 14 respiratory-condition categories at 91.59% accuracy, by building a 146-feature acoustic dataset and tuning XGBoost, MLP, Random Forest and Extra Trees models on 3,530 recordings.

Blood Cell Classification using Vision Transformer | Python, PyTorch, Score-CAM

- Achieved 99.18% test accuracy on 3 cell classes, by training a ViT from scratch on 4,886 BCCD images and interpreting predictions with Score-CAM.

EDUCATION

B.Tech, Bioinformatics | SASTRA Deemed University, Thanjavur | Aug 2022 – May 2026 | CGPA: 7.46/10

CERTIFICATIONS

- AI & Machine Learning and Data Science – Udemy, 2024
- Generative AI & Prompt Engineering – HCLTech, 2025